PERUMAL MARIMUTHU

AI Engineer
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SUMMARY

Computer Science graduate student at NYU Tandon with experience in Python, SQL, machine learning, NLP, data engineering, and AWS. Built data pipelines, automated evaluation frameworks, and AI/ML systems processing 500K+ records. Strong foundation in software engineering, machine learning, data analysis, and cloud deployment.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, SQL **ML/AI:** Machine Learning, NLP, Scikit-learn, Pandas, NumPy
Cloud & Tools: AWS EC2, AWS S3, Docker, Git, REST APIs **Data:** ETL Pipelines, Data Validation, SQL Optimization, Data Visualization

EXPERIENCE

AI Research Fellow – IndiaAI Foundation, Government of India **Oct 2024 – Aug 2025**
AI / Machine Learning

- Supported AI-focused systems and data analytics across 3 operational domains serving 100+ concurrent users using Python and SQL to monitor performance and identify trends.
- Identified usage patterns and anomalies and implemented fixes that reduced query resolution time by 40%.
- Collaborated within a 4-person cross-functional engineering, data, and operations team, contributing to 2 published research outcomes and 3 system integrations.

AI & Machine Learning Intern – AICTE & AWS Academy **Sep 2023 – Nov 2023**
Artificial Intelligence / Machine Learning

- Monitored performance and anomalies across 8+ systems, producing benchmarks and reports that contributed to a 12% average improvement across 3 projects.
- Deployed and monitored applications on AWS EC2 and S3, gaining hands-on cloud deployment experience for AI/ML services.

Data Engineer Intern – AICTE & AWS Academy **May 2023 – Aug 2023**
Data Engineering

- Designed and maintained ETL pipelines processing 500K+ records with data-quality validation for downstream systems.
- Optimized 20+ SQL queries, improving data retrieval speed by 25%.

PROJECTS

Automated Text Summarization using NLP *Python, NLP, Machine Learning*

- Built a Python-based NLP pipeline and automated evaluation framework across 10K+ records, reducing manual review time by approximately 70%.
- Implemented automated performance tracking and reporting to support iterative model evaluation and improvement.

Intelligent Assistive Eyewear – Real-Time AI System *Python, Machine Learning*

- Built and tested a real-time AI system with 15 participants and iterated through 3 improvement cycles using performance data.
- Developed dashboards enabling stakeholders to monitor system accuracy and performance.

IoT-Enabled Microclimate Monitoring System – Best Paper Award *Python, IoT, Machine Learning*

- Built automated data pipelines processing 50K+ daily records with validation and alerting, reducing data-preparation time by 45%.
- Developed a machine-learning crop recommendation workflow using environmental measurements and presented findings to 200+ international attendees.
- Received Best Paper recognition among 120+ submissions at an international IoT conference.

EDUCATION

New York University – Tandon School of Engineering **Expected May 2027**
M.S. Computer Science *Brooklyn, NY*
SRM Institute of Science and Technology **May 2025**
B.Tech. Computer Science and Engineering – GPA: 3.9/4.0 *Chennai, India*

CERTIFICATIONS

Google Foundations of Data Analysis | Machine Learning Specialization – University of Washington | Supervised Machine Learning – Stanford University | Networking & Security – Cisco